Here is a reasonable **Entity-Relationship (ER) design** based on the roles and functionalities you described earlier. This schema should cover the core requirements of a small/medium dental clinic system.

Main Entities and Relationships

Entity	Main Attributes (suggested)	Primary Key	Relationships / Notes
User	user_id, username, password_hash, email, phone, full_name, role, created_at, is_active	user_id	Super-type / base for all login accounts
Admin	(inherits from User)	(user_id)	role = 'admin'
Dentist	(inherits from User), license_number, specialization	(user_id)	role = 'dentist'
Receptionist	(inherits from User)	(user_id)	role = 'receptionist'
Patient	patient_id, first_name, last_name, date_of_birth, gender, phone, email, address, created_at, created_by (FK)	patient_id	created_by → Receptionist (User)
MedicalHistory	history_id, patient_id, recorded_date, condition_description, notes, recorded_by (FK)	history_id	recorded_by → Receptionist or Dentist
Appointment	appointment_id, patient_id, dentist_id, date_time, status, reason, notes, created_at	appointment_id	patient_id → Patient dentist_id → Dentist
TreatmentRecord	record_id, appointment_id, patient_id, dentist_id, visit_date, chief_complaint, findings, treatment_performed, next_visit_suggestion	record_id	Usually 1:1 with Appointment (or 1 appointment → many records)
Diagnosis	diagnosis_id, record_id, diagnosis_code / description, notes	diagnosis_id	Many per TreatmentRecord

Prescription	prescription_id, record_id, prescribed_date, dentist_id	prescription_id	One TreatmentRecord → many Prescriptions
Medication	prescription_item_id, prescription_id, drug_name, strength, dosage, quantity, instructions, frequency	prescription_item_id	Many per Prescription
Notification	notification_id, user_id / patient_id, type (appointment / prescription / etc.), message, sent_at, read_at, status	notification_id	Can link to Patient or any User

Text-Based ER Diagram (Crow's Foot style notation)

```text
┌──────────────┐        1         creates       0..*   ┌──────────────┐
│  Receptionist│──────────────────────────────────────│   Patient    │
└──────┬───────┘                                       └──────┬───────┘
       │                                                      │
       │ 1..* records / updates                               │ 1
       ▼                                                      ▼
┌──────────────┐        1         treats       0..*   ┌──────────────┐
│   Dentist    │──────────────────────────────────────│ Appointment  │
└──────┬───────┘                                       └──────┬───────┘
       │ 1                                                    │ 1
       │                                                      │
       ▼                                                      ▼
┌────────────────┐   1     performed_by     1..*   ┌────────────────┐
│ TreatmentRecord │◄──────────────────────────────────│     Dentist    │
└────────┬───────┘                                   └────────────────┘
         │ 1
         │
         ├─ 1..*  has ──┐
         ▼              │
┌──────────────┐        │
│   Diagnosis  │        │
└──────────────┘        │
                        │
         ├─ 1..*  generates ──┐
         ▼                    │
```

```
┌──────────────────┐                 │
│  Prescription    │◄────────────────┘
└──────────┬───────┘
           │ 1
           ▼
┌──────────────────┐
│ PrescriptionItem  │  (drug, dose, qty, instructions …)
└──────────────────┘

┌──────────────┐        1         belongs_to      *    ┌──────────────┐
│   Patient    │──────────────────────────────────────│ MedicalHistory│
└──────────────┘                                        └──────────────┘

┌──────────────┐        1         receives       0..*   ┌──────────────┐
│   Patient    │──────────────────────────────────────│  Notification │
└──────────────┘                                        └──────────────┘
       ▲ 1
       │
       │ 0..*
┌──────────────┐
│   User       │  (can also receive notifications — dentists, receptionists)
└──────────────┘
```

Quick Relational Summary (most common foreign keys)

- **Patient.created_by** → User (usually receptionist)
- **Appointment.patient_id** → Patient
- **Appointment.dentist_id** → User (role = dentist)
- **TreatmentRecord.appointment_id** → Appointment
- **TreatmentRecord.dentist_id** → User (role = dentist)
- **TreatmentRecord.patient_id** → Patient
- **Diagnosis.record_id** → TreatmentRecord
- **Prescription.record_id** → TreatmentRecord
- **Prescription.dentist_id** → User (role = dentist)
- **PrescriptionItem.prescription_id** → Prescription
- **Notification.patient_id** → Patient (nullable)
- **Notification.user_id** → User (nullable)

Implementation Notes / Choices You Can Adjust

- You can merge **Diagnosis** and **TreatmentRecord** if you want simpler design
- Some clinics keep **Prescription** directly under **TreatmentRecord** (without separate entity)
- You may want to add **Tooth / Quadrant / Surface** information for dental-specific charting
- Consider adding **Clinic / Branch** entity if you plan multi-location support later
- Add **Audit / Log** table if compliance (HIPAA-like) is important
- For notifications: you can use a simple polymorphic approach (patient_id OR user_id) or separate tables

Would you like me to give you:

1. SQL CREATE TABLE statements for PostgreSQL / MySQL ?
2. A more detailed version with data types & constraints ?
3. A simplified version (fewer tables) ?
4. A version with dental charting / tooth-level data ?

Just let me know which direction you want to go next.